

REVIEW | *Recovery from Exercise*

Recovery of central and peripheral neuromuscular fatigue after exercise

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Carroll TJ, Taylor JL, Gandevia SC. Recovery of central and peripheral neuromuscular fatigue after exercise. *J Appl Physiol* 122: 1068–1076, 2017. First published December 8, 2016; doi:10.1152/jappphysiol.00775.2016.—Sustained physical exercise leads to a reduced capacity to produce voluntary force that typically outlasts the exercise bout. This “fatigue” can be due both to impaired muscle function, termed “peripheral fatigue,” and a reduction in the capacity of the central nervous system to activate muscles, termed “central fatigue.” In this review we consider the factors that determine the *recovery* of voluntary force generating capacity after various types of exercise. After brief, high-intensity exercise there is typically a rapid restitution of force that is due to recovery of central fatigue (typically within 2 min) and aspects of peripheral fatigue associated with excitation-contraction coupling and reperfusion of muscles (typically within 3–5 min). Complete recovery of muscle function may be incomplete for some hours, however, due to prolonged impairment in intracellular Ca^{2+} release or sensitivity. After low-intensity exercise of long duration, voluntary force typically shows rapid, partial, recovery within the first few minutes, due largely to recovery of the central, neural component. However, the ability to voluntarily activate muscles may not recover completely within 30 min after exercise. Recovery of peripheral fatigue contributes comparatively little to the fast initial force restitution and is typically incomplete for at least 20–30 min. Work remains to identify what factors underlie the prolonged central fatigue that usually accompanies long-duration single joint and locomotor exercise and to document how the time course of neuromuscular recovery is affected by exercise intensity and duration in locomotor exercise. Such information could be useful to enhance rehabilitation and sports performance.

central fatigue; endurance, exercise; muscle fatigue; recovery

SUSTAINED PHYSICAL EXERCISE leads inexorably to a reduced capacity to produce voluntary force. Although multiple processes contribute to this “muscle fatigue,” it is ultimately manifest as impaired muscle function and/or a reduction in the capacity of the central nervous system to activate muscles. The term “peripheral fatigue” is typically used to describe force reductions due to processes distal to the neuromuscular junction, whereas those due to processes within motoneurons and the central nervous system are commonly known as “central fatigue.” The physiology of fatigue has been studied for well over a century (see Ref. 28 for a comprehensive historical review), and recent reviews have summarized current understanding of various aspects of fatigue (2, 13, 23, 62, 84, 92, 93). As part of this Highlighted Topic series, we consider the factors that determine the *recovery* of voluntary force-generating capacity after various types of exercise.

In attempting to document the mechanisms of fatigue and recovery, an important consideration is that sustained exercise

affects physiological processes throughout the neuromuscular system. Critically, alterations in these underlying processes may either contribute to, or compensate for, fatigue (see sections below for examples and details). Although some progress has been made in documenting interrelationships between exercise characteristics, physiological responses, and impaired force-generating capacity, much remains to be learned. Given this, and the constraints of the review format, we consider primarily recovery of neuromuscular performance in terms of voluntary or artificially evoked forces measured in intact humans. We attempt to link these functional measures of fatigue with the likely underlying processes where possible and highlight areas in which a lack of available evidence prevents this. As a consequence of our focus on work that we believe provides the clearest inferences regarding the mechanisms of recovery, a number of interesting and important issues are omitted. For example, we do not consider the effects of low-intensity exercise, nutrition, or other interventions on the recovery process. Some of these issues are dealt with in other papers in the Highlighted Topic series.

Because of the task-dependent nature of fatigue (23), the review is structured according to sections that each consider

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the classes of exercise in which recovery has been documented. We initially consider maximal voluntary contractions at a single joint, because central fatigue is most easily studied in this type of task. In particular, we extrapolate recent insights into the mechanisms of central fatigue in these tasks to the postexercise recovery period. We then consider what general implications can be drawn about recovery from apparent differences in fatigue and recovery between maximal and sustained submaximal contractions at a single joint. Finally, we consider recovery from everyday exercise, such as running and cycling, which involve large muscle masses, and consequently challenge systemic homeostasis.

Recovery from Maximal Contractions

A reduction in the maximal force that a person can produce during a isometric maximal voluntary contraction (MVC) provides the most straightforward demonstration of fatigue. Accordingly, tasks involving a sustained MVC provide a convenient model to study fatigue and recovery, because there is a continuous measure of fatigue during the protocol (i.e., instantaneous MVC force) and because recovery can easily be tracked with the same apparatus used to induce fatigue (i.e., without the requirement to reposition the subject or switch between tasks). Isometric conditions are also convenient for measurement of forces evoked artificially by electrical or magnetic stimulation of motor nerves, descending tracts, or the motor cortex. Evoked forces at rest can provide information about fatigue and recovery of the muscle fibers, whereas force responses to stimulation that is “superimposed” upon voluntary contractions can reveal the extent to which voluntary neural drive is sufficient to generate the maximal force of which the muscles are capable.

It is important to acknowledge that, although measures obtained during MVC provide valid and easily interpreted information about neuromuscular function, such measures may not be ideally sensitive to some physiological changes that are important for exercise performance. Examples that demonstrate this point include observations of exacerbated force declines during low-frequency motor unit firing (16, 22, 83, 95, 102) and observations that some interventions such as hyperthermia (98) or prior locomotor exercise (83) have much greater effects on sustained than brief MVC performance. Despite these limitations, in many cases measurements obtained during MVC provide the best available evidence regarding muscle force-generating capacity and the capacity of the central nervous system to drive muscles, and the current review will focus extensively on work that exploits these measures.

Maximal voluntary force declines rapidly and progressively during a sustained MVC, typically falling to below 50% of baseline within 1–2 min. There is also a rapid but partial *recovery* of voluntary force over the first few minutes after cessation of this type of exercise, with the largest component occurring within 15–30 s. This suggests that reperfusion of the exercising muscles is a key factor in initial recovery, a conclusion supported by the observation that recovery is delayed if the muscles are held ischemic. Further recovery of MVC force is much slower and may reach only ~80% by 4–5 min postexercise (see Fig. 1A and Refs. 29, 50, 98).

The size of superimposed twitches evoked by stimulation of motor nerves or the motor cortex also increases within 15–30 s

of sustained MVC, indicating that part of the voluntary force reduction is due to suboptimal output from the motor cortex (29, 40, 41, 48, 50, 51, 91, 98). This failure of voluntary activation has been estimated to account for ~25% (100) of the total force reduction during sustained maximal contractions, but voluntary activation usually completely recovers to pre-fatigue levels within ~30 s of exercise termination (see Fig. 1C and Refs. 29, 48, 50, 51, 98). The dissociation in the time course of recovery between MVC and voluntary activation implies that the sustained impairments in voluntary force production originate predominantly within the muscle fibers. Further support for this conclusion derives from observations of prolonged, incomplete recovery of electrically evoked twitches and tetani after repeated isometric contractions to the limit of tolerance (22).

Mechanisms of Recovery

A detailed coverage of what is currently known about the physiological processes that accompany sustained exercise is beyond the scope of this paper, but see Taylor et al. (92) and Allen et al. (2) for fuller accounts of central and peripheral fatigue mechanisms, respectively. Here, we provide a brief overview (see Fig. 2 for a summary) and emphasize that the time courses of change in these processes need not reflect that of the functional recovery in voluntary force. This is because physiological responses to sustained exercise may either contribute to, or compensate for fatigue, and recovery of voluntary force is ultimately determined by the interplay of such underlying processes. For example, during sustained maximal contractions, both the excitatory and the inhibitory (silent period) responses of motor cortex output cells to transcranial magnetic stimulation increase. These changes suggest extra cortical excitability, which should improve motor output, but also extra cortical inhibition, which might contribute to fatigue. At the same time the extent to which voluntary output from the cortex can harness the full capacity of muscles decreases (i.e., there is supraspinal fatigue; e.g., Refs. 29, 40, 41, 91). Stimulation during intermittent MVCs with different duty cycles shows that these three effects have different time courses of development and return to baseline, with the silent period returning to baseline in ~10 s, the excitatory response to cortical stimulation in 15–30 s, and supraspinal fatigue in ~1 min (91). Although the factors that underlie a failure to harness the full capacity of cortical outputs to drive motoneurons appropriately for maximal voluntary force generation are not known, a role for feedback from group III and IV muscle afferents is likely.

When firing of metabolically sensitive muscle afferents is prolonged after a fatiguing contraction by preventing blood flow to the muscle, supraspinal fatigue continues until blood flow is allowed to resume (29, 51). Moreover, firing of afferents from the fatigued muscle affects voluntary activation of other muscles in the same limb (50, 51). In contrast, the excitatory and inhibitory responses elicited by stimulation of the motor cortex typically return to pre-exercise values despite the occlusion (29, 51). This suggests that muscle afferent firing may limit drives to the motor cortex (and other descending) output cells during maximal effort, without apparent direct actions on motor cortical cells. However, debate continues on the actions of group III and IV afferents on motor cortical excitability because responses evoked by stimulation of the

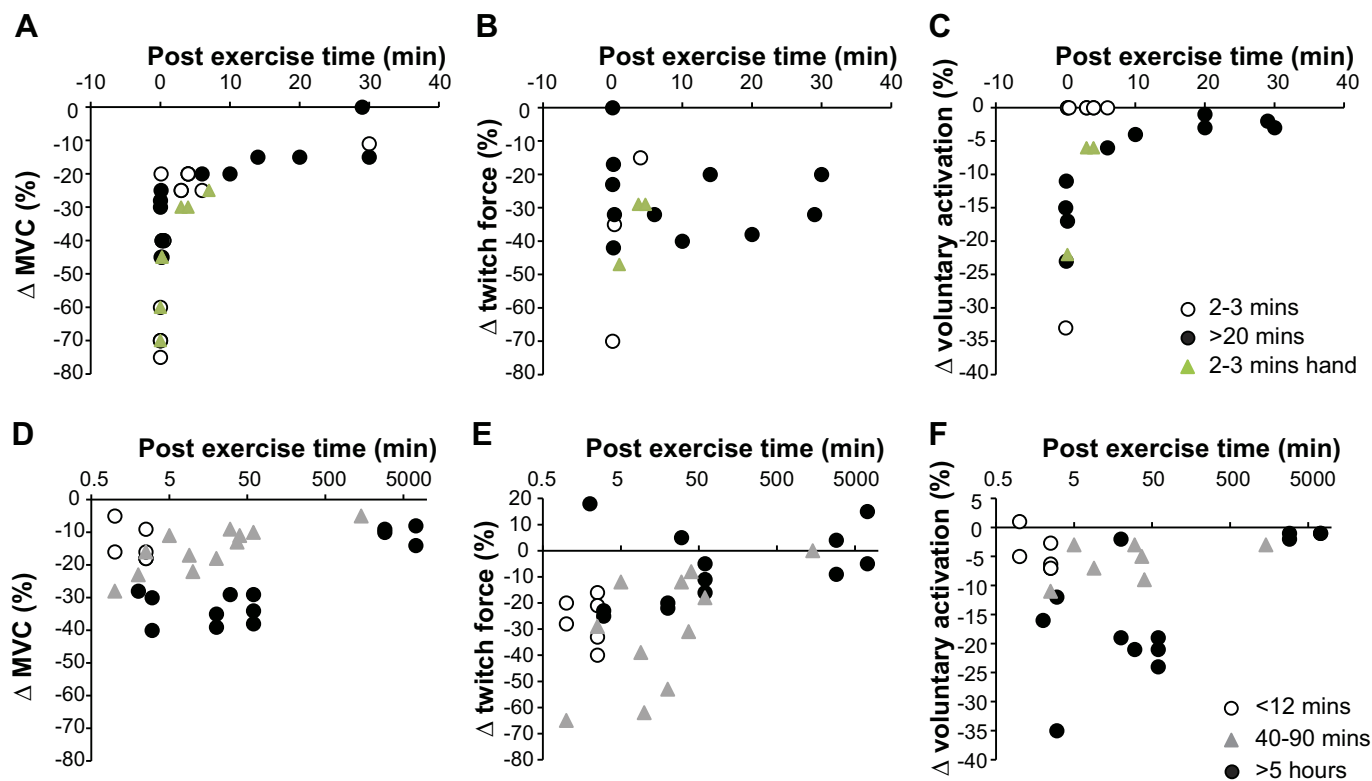


Fig. 1. Percentage change in MVC (A, D), twitch amplitude (B, E), and voluntary activation (C, F) measured at various times after the cessation of fatiguing exercise. A–C show data from isometric contractions; D–F show data from locomotor exercise. Values are taken from text or estimated from figures across a number of papers. Some papers contribute multiple points and others only one. No distinction was made between voluntary activation measured with motor nerve or motor cortex stimulation. Note that the time postexercise in minutes is on a logarithmic scale for the locomotor exercise only. Papers contributing to the graphs for isometric contractions of 2–3 min (open circles) are (29, 32, 48, 51, 98). For 2–3 min contractions of hand muscles (green triangles), contributors are (50, 76). For contractions of >20 min (solid circles), contributing papers are (88, 89, 102, 106). Apart from the hand muscles, data are from elbow flexors and knee extensors. Papers contributing to the locomotor exercise graphs for exercise of <12 min (open circles) are (33, 35, 86, 96, 97). Contributing papers for exercise of 40–90 min (gray triangles) are (16, 74, 75, 83). For exercise of >5 h (solid circles), papers are (58, 65, 72, 94, 95). Exercise of <12 min represents only knee extensors, whereas both longer durations also contain some plantarflexor data. Also of note, the exercise of >5 h was running, 40–90 min included cycling, running, and a soccer game, and exercise of <12 min included cycling and one study of running.

cortex and measured in the muscle are influenced by both cortical and spinal excitability. Hence interpretation of changes in responses to cortical stimulation is not clear cut (49). Indeed, it is possible that supraspinal fatigue could occur despite relatively stable outputs from supraspinal centers. Here, central fatigue would be generated by changes in input-output properties of the motoneuron pool, such that a similar set of cortical outputs that are untapped by volition and available to artificial stimulation would have a proportionally greater effect on muscle force.

In contrast to the uncertainty regarding supraspinal contributions to fatigue, it is clear that central fatigue must be affected by the motoneuron pool itself (see Refs. 28, 57, 92 for reviews). Changes at this level can arise from tonic and phasic reflex inputs and other inputs associated with the exercise as well as changes in intrinsic properties of the motoneurons. Superimposed on such changes are neuromodulatory effects, produced for example, by descending monoaminergic drives. Although these changes are the focus of current work in human and animal studies, it is not simple to link their effect to a precise aspect of motoneuronal or spinal behavior or to determine their effect on motor output in a voluntary contraction. However, two things are clear. First, changes in the excitability of the motoneuron pool must be compensated by changes in

descending drive to keep motoneuronal output constant. Hence a reduction in excitability (through inhibition or disfacilitation, see below) would necessitate greater drive. Such a reduction would likely produce a greater subjective effort for the same submaximal motor output. Second, the changes documented so far at a motoneuronal level have a range of time courses, ranging from milliseconds to minutes. Some examples are given briefly below.

Inputs from group III/IV muscle afferents can act at segmental sites to modify excitability of the motoneurons and at supraspinal sites to affect the level of drive to the motoneuron pools (13, 18, 28, 57, 85). Existence of these effects has long been studied with circulatory occlusion (e.g., Ref. 14). Not surprisingly, restoration of muscle blood flow and removal of K^+ and other metabolites rapidly attenuates the central effects of group III/IV muscle afferent firing with recovery of voluntary activation in ~30 s (48, 50, 51). More recently, lumbar intrathecal injection of fentanyl has been used to reduce group III/IV inputs to the central nervous system and attenuate an inhibition of voluntary motor output (e.g., Refs. 3, 5).

Two approaches illustrate depression of motoneuronal “excitability” after voluntary isometric exercise. First, during relaxation after contraction, the propensity of the motoneurons to discharge a recurrent action potential (termed an F wave) is

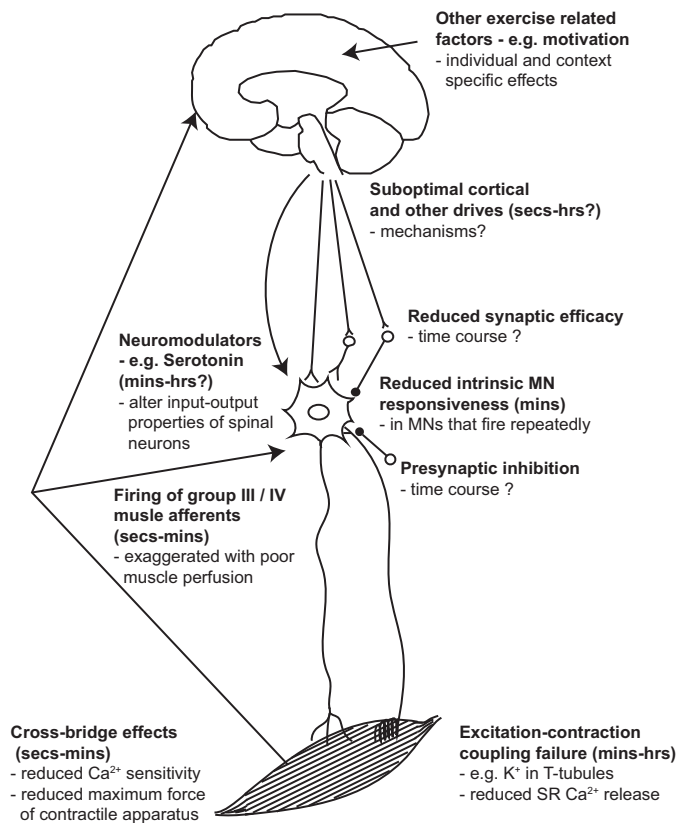


Fig. 2. Schematic illustration highlighting some key neuromuscular responses to exercise. Various physiological processes that are affected by fatigue are identified in bold lettering, with the approximate time course of their recovery (or return to baseline) indicated in parentheses. Additional information about the mechanism by which physiological changes impair function is provided where possible. MN, motoneuron.

depressed for several minutes after a 2-min MVC (e.g., Refs. 52, 81). This depression occurs in hand and leg muscles and is less for weaker contractions (53). Although in simple terms this can be considered a depression in *intrinsic* motoneuronal behavior (i.e., because it is seen in an evoked response that does not require synaptic activation), one constraint is that the measurement is dominated by changes in large high-threshold motoneurons in the pool (e.g., Refs. 24).

Second, evidence for profound change at a spinal level comes from the use of high-intensity conditioning TMS during an MVC to interrupt descending voluntary drive and allow the motoneurons to be tested during artificial “relaxation.” Testing is done with a cervicomedullary stimulus, which produces a muscle response by activation of corticospinal axons. Studying motoneuron behavior in the absence of volitional activity greatly simplifies the range of factors that are at play and makes it possible to determine mechanisms. After 15 s of an MVC of elbow flexors, the corticospinal response is virtually abolished (61). This spinal inhibition affecting the corticomotoneuronal path takes 2–3 min to recover after the end of the fatiguing MVC. The phenomenon also occurs during and after submaximal contractions and preferentially affects the motoneurons active in the contraction (59).

Finally, although detailed consideration of the intramuscular processes that determine recovery from exercise are beyond our scope (refer to Ref. 2), characteristics of evoked forces

illustrate some general principles. For example, reductions in evoked twitch magnitude and tetanic forces evoked by low-frequency stimulation are consistently greater than declines in MVC or high-frequency tetanic force (16, 22, 83, 95, 102). The time course of recovery of forces evoked by high-frequency stimulation is also much more rapid than that of recovery of low-frequency stimulation forces (or twitches). Force produced by high-frequency stimulation returns near to baseline within 20 min, even after a prolonged series of contractions to the limit of endurance in the presence of ischemia, whereas low-frequency force impairments can persist for more than 24 h (22). Differential fatigue and recovery effects as a function of motoneuron firing frequency likely follow from the sigmoidal shape of the Ca^{2+} force relation (2) and may reflect alterations in release or reuptake of Ca^{2+} from the sarcoplasmic reticulum or reduced Ca^{2+} sensitivity at the contractile apparatus. Note that single twitches create conditions that lie close to the origin of the Ca^{2+} force relation. By contrast, the rapid partial restitution of high-frequency force in the first seconds of recovery probably follows from muscle reperfusion, with clearance of K^{+} allowing repolarization of the t-tubule membranes likely to play a major role (2).

Moreover, the general principle that responses to sustained exercise can either contribute to, or compensate for, fatigue holds for peripheral as well as central processes. For example, exercise can cause a slowing in the contractile properties of muscle, such that a lower rate of muscle fiber action potentials is required to generate a fully fused tetanus (39, 100, 103). This type of effect would partially compensate centrally mediated declines in motoneuronal firing rates, although the presence of central fatigue underscores the fact that, despite this partial compensation, voluntary drive is insufficient to generate the maximum evocable muscle force.

Recovery from Sustained Submaximal Contractions

The contrast between the fatigue responses in maximal and sustained or intermittent submaximal contractions can inform understanding of the factors that determine recovery after exercise (93). Although central fatigue cannot be measured using peripheral or cortical stimulation *during* the submaximal task, it can be documented during maximal efforts inserted within the main task. Furthermore, although not a direct measure of central fatigue, perceived effort increases out of proportion to the level of EMG. This is best seen in a sustained contraction in which the participant holds a submaximal target EMG level. In such contractions, the alteration in the EMG to force relationship produced by peripheral fatigue results in reduced force output. However, participants report that progressively more effort is required to do the task; which is to produce the same EMG (e.g., Ref. 60). This suggests that central mechanisms also influence performance *during* submaximal tasks (56, 80, 81). A key distinction between maximal and submaximal tasks is that additional motor units are progressively recruited as fatigue develops during sustained low-force contractions (1, 19, 30). By contrast, it is likely that all available motor units are recruited at high rates at the beginning of a sustained MVC and firing rates progressively decline with fatigue and may eventually cease in some high-threshold units (e.g., Ref. 71). Thus, for a given contraction duration, less fatigue occurs in high-threshold units for submaximal than

maximal contractions. This may be related to the observation that central fatigue contributes proportionally more to the total force reduction during sustained submaximal than maximal contractions. For example, impaired voluntary activation accounts for ~65% of the reduced MVC during 70 min of elbow flexion at 5% MVC (88), ~40% of the MVC drop during 43 min of contraction at 15% MVC (89), but only ~25% of the force drop for a 2 min MVC (100).

There is likely to be some maintenance of muscle perfusion during submaximal contractions, depending on the target force, and duty cycle when contractions are intermittent, which should reduce the accumulation of metabolites that leads to both firing of the subset of group III and IV afferents that are sensitive to noxious stimuli and t-tubule depolarization by K^+ . Accordingly, resting twitches evoked by motor nerve stimulation do not recover appreciably within 20–30 min after sustained, weak contractions of the elbow flexors (88, 89) (see Fig. 1B), suggesting that mechanisms of peripheral fatigue in such conditions relate mainly to impaired intracellular Ca^{2+} handling or sensitivity.

Despite slow recovery of evoked twitch forces, MVC force typically shows rapid, but partial, recovery within the first few minutes after termination of sustained submaximal contractions. Voluntary activation measured by motor nerve or motor cortical stimulation has a correspondingly rapid initial recovery component but may not return to prefatigue levels until 20–30 min postexercise (46, 47, 88, 89, 104, 105) (Fig. 1C). Perceived effort, measured during brief efforts, takes ~5 min to recover fully (80) but has not often been documented. Thus,

although the initial, partial restoration of voluntary force after sustained low force contractions is likely due to central fatigue recovery, impaired voluntary activation persists for longer after submaximal contractions sustained for 6–70 min than after maximal contractions sustained for up to 2 min. The mechanism underlying this delayed central recovery is not known.

Recovery from Locomotor Exercise

Sustained contractions at a single joint are a convenient model to study fatigue and involve physical demands that are similar to some activities of daily living (e.g., holding a bag of groceries). However, there is uncertainty about the degree to which the processes that constitute fatigue in such tasks also apply to activities such as walking, running, and cycling, which typically require higher rates of energy use, and consequently greater cardiovascular and ventilatory demands. There is an extensive literature on the physiological responses to fatiguing locomotor exercise (see Refs. 37, 62, 66, 84 for reviews), but direct measurement of muscle fatigue is challenging in such tasks because it is difficult to measure force-generating capacity during and immediately after exercise: there is typically some delay required to couple subjects to a myograph and initiate neuromuscular recording and stimulation. Nonetheless, muscle fatigue has been documented after running (55, 58, 65, 78, 80, 95, 101), cycling (4, 5, 7–12, 15, 20, 33, 35, 38, 43, 44, 54, 56, 67, 68, 79, 83, 85, 94, 96, 97), and skiing (63) of durations ranging from a few minutes to multiday ultraendurance events (see also Refs. 13, 62 for review). Care is needed

Table 1. Summary of central and peripheral fatigue responses to exercise and recovery, categorized by modality, duration, and intensity of exercise

Peripheral Fatigue	Central Fatigue
Exercise Modality	
Sustained MVC metabolite accumulation (e.g., K^+) fast recovery with re-perfusion (<30 s)	Sustained MVC ↓ VA due to group III/IV firing fast recovery with reperfusion (<90 s)
Intermittent Shortening/Isometric less blood occlusion and K^+ build-up slower recovery due to $[Ca^{2+}]_i$ effects (min to h)	Intermittent Shortening/Isometric ↓ VA and recovery time course depends on exercise duration can vary from 1 to 30+ min
Lengthening damages muscle fibers slow recovery (days to weeks)	Lengthening muscle damage leads to ↓ VA* slow recovery (days)
Exercise Duration	
Short Duration (<2–3 min) metabolite accumulation crucial (e.g., K^+) fast recovery with reperfusion (<30 s)	Short Duration (<2–3 min) short lasting ↓ VA (<90 s) ↓ MN responsiveness (min)
Long Duration (>6 min) recovery depends on number of high $[Ca^{2+}]_i$ episodes (intensity and duration) $[Ca^{2+}]_i$ effects can last min to h glycogen depletion possible (h)	Long Duration (>6 min) recovery appears closely related to duration (mechanisms unknown) can vary from 1 to 30+ min
Exercise Intensity	
High (near MVC/sprints) all muscle fibers recruited metabolite accumulation crucial (e.g., K^+) fast recovery with reperfusion (<30 s)	High (near MVC/sprints) ↓ VA due to group III/IV firing fast recovery with reperfusion (<90 s) responsiveness of all MNs reduced
Low (low forces/endurance exercise) low threshold, fatigue-resistant units recovery depends on duration	Low (low forces/endurance exercise) ↓ VA occurs if duration long recovery can vary from 1 to 30+ min low threshold MN excitability ↓ (min)

VA, voluntary activation; MN, motoneuron. *Note that VA was reduced for more than 1 day after eccentric exercise when measured with motor nerve stimulation, but not motor cortical stimulation (73). The reason for this discrepancy is not clear, but it illustrates that different measures of VA do not always respond in the same way during fatigue and recovery.

in interpretation of this literature, however, because time and logistical constraints sometimes prevent satisfaction of criteria necessary to ensure valid measurements (see Refs. 28, 99). Note also that general trends in recovery time courses are more difficult to identify from the available data on locomotor exercise (Fig. 1).

During locomotor exercise at a constant power output, sense of effort and EMG amplitude increase progressively over time (9–11, 35, 97), suggesting that fatigue accumulates throughout exercise. Although it is difficult to measure muscle fatigue directly within the first 1–2 min postexercise, evidence from a rhythmic “locomotor-like” knee flexion/extension task suggests that there is rapid, but partial, recovery over tens of seconds that is typical of sustained maximal and submaximal isometric contractions (26, 27, 67). However voluntary force capacity is still reduced from baseline at 1–3 min after termination of fatiguing locomotor exercise, and this is due to both peripheral and central fatigue (7, 33, 35, 38, 83, 96, 97). As for single joint isometric contractions, the relative contribution of impaired muscle function and voluntary activation to muscle fatigue probably depends upon the duration and intensity of exercise (17, 96, 97), with peripheral fatigue contributing relatively more to MVC reduction after short, high-intensity exercise, and central fatigue contributing relatively more during longer-duration, moderate-intensity exercise (see Fig. 1, D–F). The extent of central fatigue development may depend more on exercise duration than intensity, because longer-duration trials resulted in greater voluntary activation reductions than short-duration trials when exercise was self-paced and involved a high-intensity “end-spurt” (97).

Both central and peripheral fatigue can persist for well over 30 min after prolonged locomotor exercise, with extreme endurance events lasting many hours or days reportedly resulting in the longest-lasting impairments (72, 79, 80, 83, 95). Repeated sprints and sports such as tennis and soccer also induce prolonged central and peripheral fatigue (31, 34, 42, 66, 69, 70, 75). However, systematic attempts to document the time course of recovery as a function of exercise duration and/or intensity have not been made. Recovery is further complicated for running, which involves eccentric contractions that induce muscle damage, because damage induces long-lasting impairments in evoked muscle forces and voluntary activation (73). Despite this, it appears that the determinants of peripheral fatigue recovery may be similar for single joint isometric contractions and locomotor exercise. Prolonged reductions in twitch forces and forces evoked by low-frequency stimulation occur in both (28, 56, 62, 83, 102), as do reductions in Ca^{2+} ATP-ase activity and Ca^{2+} uptake into the sarcoplasmic reticulum (16, 22, 32, 102). By contrast, central fatigue persists longer after locomotor exercise than after high-force isometric contractions. The factors that underlie persistent central fatigue after sustained low-force isometric contractions might be important, but there is also the possibility that factors associated with homeostatic regulation of body temperature (33, 68, 77, 98), systemic oxygen or carbon dioxide concentrations (6, 11, 12, 21, 36, 45, 64, 77, 82, 87), and metabolism (90) contribute. One potential candidate is reduced serotonergic or other neuromodulatory inputs to the motoneuron pool. For example, 35–40 min of fatiguing locomotion reduces firing rates of serotonergic neurons in the medullary raphe nuclei of

cats (25), and spontaneous firing takes ~45 min to recover to baseline rates.

Conclusions

The time course and mechanisms of recovery after fatiguing exercise are highly dependent on the characteristics of the preceding exercise bout. We have summarized some key points in Table 1, but considerable work remains before a complete description of the recovery processes will be possible. It is currently unclear what factors underlie the prolonged central fatigue that can accompany long-duration single joint and locomotor exercise. Work also remains to document how the time course of neuromuscular recovery is affected by exercise intensity and duration in locomotor exercise. Better understanding of the factors that modulate recovery from muscle fatigue may be of practical use to enhance rehabilitation and sports performance.

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DISCLOSURES

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AUTHOR CONTRIBUTIONS

T.J.C. and J.L.T. prepared figures; T.J.C., J.L.T., and S.C.G. drafted manuscript; T.J.C., J.L.T., and S.C.G. edited and revised manuscript; T.J.C., J.L.T., and S.C.G. approved final version of manuscript.

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